CM5 GUARD A — ENGINEERING REVIEW ONLY — DO NOT FABRICATE / NO MOTOR OR TX OPERATION

Every net is labelled. NC contacts to GND: open wire inhibits hardware; external DC E–stop is mandatory.

Powerpole / EMI / isolated power / CM5IO PD / RF bulkhead SMA stay EXTERNAL, unresolved ratings.

J1
CM5IO GPIO VIO=3V3 ONLY
I2C1_SDA
I2C1_SCL
AZ_LIMIT_MIN_N
GND_LOGIC
SPI1_CS1_EL
GNSS_PPS
PTT_VHF_REQUEST
ESTOP_AUX_N
POL_LIMIT_MAX_N
WATCHDOG_HEARTBEAT
AZ_LIMIT_MAX_N
EL_LIMIT_MIN_N
TCA9548A_RESET_N
SPI1_MISO
FAULT_POL_N
GNSS_UART_TX
GNSS_UART_RX
SPI1_CS0_AZ
PTT_UHF_REQUEST
FAULT_AZ_N
FAULT_EL_N
POL_LIMIT_MIN_N
EL_LIMIT_MAX_N
MOTOR_ARM_REQUEST
SPI1_CS2_POL
SPI1_MOSI
SPI1_SCLK
J2
ISOLATED 5V external PSU
AUX_5V_ISOLATED
U1
AP2112K–3.3TRG1
VIN
VOUT
EN
GND
V3V3
C1
BAT_POS
GND_MOTOR
BAT_FUSED
POWER_GATE
BAT_PROTECTED
DC–RATED EXTERNAL ESTOP OUT
EXTERNAL QUALIFIED USB–C PD CONVERTER
VHF PA external
MOTOR_BRANCH_FUSED
CM5IO_BRANCH_FUSED
VHF_BRANCH_FUSED

POWER CANDIDATES DNP: empty fuseholders, unqualified ratings; no motor/PA/battery operation authorized.

Manual rearm latch requires healthy NC/E–stop/watchdog and arm/PTT requests LOW at local button edge.

SMA pairs are passive feed–through ONLY. No PA/filter/RHCP–LHCP integration or RF validation claimed.

DO NOT FABRICATE — POWER QUALIFICATION PENDING

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Title: CM5 Guard AZ/EL/POL

Size: A0	Date:	**Rev: A review only**
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